$(a)$

| $\hat{\theta}_i$ | $\lambda_i/\lambda_1$ | $\theta_1$ | $\theta_2$ | $\theta_3$ |
|---|---|---|---|---|
| 1 | 1.0000 | 0.1 | 0.3 | 1.0 |
| 2 | 0.0097 | 1.0 | 0.7 | −0.3 |
| 3 | 0.0010 | −0.8 | 1.0 | −0.2 |